

The  $pK_a$  values of the functional groups of lysine are shown in the following table.

| Functional group   | $pK_a$ |
|--------------------|--------|
| $\alpha$ -carboxyl | 2.18   |
| $\alpha$ -amine    | 8.95   |
| Side chain amine   | 10.53  |

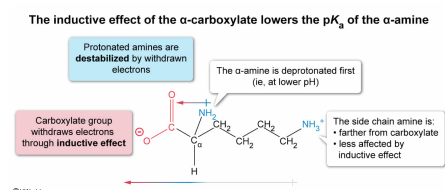
Which of the following statements best explains why the  $pK_a$  value of the  $\alpha$ -amine is lower than the  $pK_a$  value of the side chain amine?

- ☐ A. The  $\alpha$ -carboxylate stabilizes the  $\alpha$ -amine through charge-charge interactions. [10%]
- ☒ B. The  $\alpha$ -carboxylate withdraws electrons from the  $\alpha$ -amine due to the inductive effect. [53%]
- ☐ C. The protonated  $\alpha$ -amine is stabilized through resonance. [18%]
- ☐ D. Different resonance forms stabilize the deprotonated amine. [17%]

Correct

52% answered correctly

### Explanation:



The  $pK_a$  value of an ionizable functional group is the pH value at which the protonated and deprotonated forms of the functional group are present in equal concentrations. As such, the  $pK_a$  value is influenced by local environmental and structural factors that affect the stabilities of the protonated vs. deprotonated forms. If the protonated form is stabilized, the functional group resists deprotonation until higher pH values, raising the  $pK_a$ . Similarly, destabilization of the deprotonated form allows the group to pick up a proton more easily, raising the pH at which significant protonation can occur, and thus raising the  $pK_a$ .

The inductive effect occurs when one or more functional groups donate or withdraw electron density from nearby ionizable groups. This effect works through covalent bonds. By withdrawing electrons, the electronegative oxygens of a carboxylate ion stabilize negative charges and destabilize positive charges of nearby functional groups. Thus, the  $\alpha$ -carboxylate destabilizes the positively charged  $\alpha$ -amine. In contrast, the side chain amine is farther away from the  $\alpha$ -carboxylate and less affected by the inductive effect. Destabilization of the protonated  $\alpha$ -amine means the  $\alpha$ -amine gives up its proton more easily at a lower pH, and therefore **the  $\alpha$ -amine has a lower  $pK_a$  due to the inductive effect.**

**(Choice A)** Charge-charge stabilization of the positive, protonated  $\alpha$ -amine would tend to *increase* its  $pK_a$  by making the group resistant to deprotonation.

**(Choices C and D)** The  $\alpha$ -amine of an isolated amino acid like lysine does *not* participate in resonance. The  $\alpha$ -amine is connected only to the  $sp^3$  hybridized  $\alpha$ -carbon, which does not have any  $\pi$ -bonds to allow for delocalized electrons.

### Educational objective:

The  $\alpha$ -connected functional groups of amino acids have lower  $pK_a$  values than similar side-chain functional groups. This variation in  $pK_a$  value is due to effects such as the inductive effect, through which nearby electronegative atoms destabilize protonated functional groups and stabilize deprotonated groups.

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